
Porter Cook

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SUMMARY

Mechatronics engineer and Visiting Instructor of Practice at Utah Tech University with industry experience in automated test equipment, FANUC, Yaskawa, and Universal Robots programming, PLC-to-robot integration, and C#/SQL manufacturing software.

KEY SKILLS

Industrial Robot Programming, PLC Programming, PLC-to-Robot Integration, Automated Test Equipment, Control Panel Design & Industrial Wiring, HMI Design, Troubleshooting & Commissioning, Curriculum Development, Technical Instruction, Lab Safety

TECHNICAL SKILLS

- PLC & Controls: Ladder Logic, Structured Text, Productivity Suite, Beckhoff TwinCAT 3, DirectLogic, SCADA, VFDs, Group I/O, Ethernet Communication
- Robotics: FANUC (HandlingTool, RoboGuide), Yaskawa, Universal Robots, End-of-Arm Tooling
- Software: C#, WPF, SQL, Python, C++, HTML
- CAD / CAM: SolidWorks, Fusion 360, Onshape, AutoCAD Electrical, CNC Toolpaths & G-code

PROFESSIONAL EXPERIENCE

Visiting Instructor of Practice, Industrial Robotics

Apr '26 - Present

Utah Tech University

Saint George, UT

- Co-developed a new Industrial Robotics curriculum, adding robot motion principles (static and inertial forces), robot I/O, and Ethernet communication alongside robot programming and operation
- First instructor to bring FANUC RoboGuide into the course; students build, program, and debug virtual workcells before running physical robots, making hardware lab time more productive
- Co-designed a PLC-to-robot trainer pairing an AutomationDirect P1000 PLC with a FANUC LR Mate 200iD
 - Wired through CRMA58 terminal blocks for hands-on group I/O, digital logic, and Ethernet labs
- Established lab safety protocols for student operation of industrial robots; teach MTRN 2200 and MTRN 2205 covering safe industrial wiring and troubleshooting

Automation Technician

Mar '26 - Apr '26

IME Automation

- Supported an automated system project from development through completion, handling controls integration and troubleshooting with end-of-arm tooling and robotic material handling

Automation Engineering Intern

Aug '24 - Sep '25

Baxter Aerospace

- Developed HVX, an automated wire-harness test system for Quality Assurance
 - Verifies continuity and performs high-voltage testing using PLC control and custom test sequences
- Built C# WPF applications backed by SQL databases to deliver MES work instructions to assembly lines
- Programmed Yaskawa, Universal Robots, and FANUC robots to automate manufacturing processes; integrated PLC control for sensor data collection and contributed to R&D on prototype automation projects

Makerspace Intern

Aug '22 - May '23

Utah Tech University - Innovation Plaza

Saint George, UT

- Lead operator for the Stinger CNC router (Fusion 360 toolpaths, G-code); installed E-stops and circuit breakers on shop machinery; trained users on laser engravers, CNC, and 3D printers

PROJECTS

Lovesac Material Dispensing System - Senior Capstone

Industry Partner

Utah Tech University

Saint George, UT

- Designed and wired the main control enclosure (AC/24 VDC distribution, PLC I/O, dual VFDs with braking resistors) and wrote the ladder logic and HMI for the dispensing sequence
- Trained polynomial-regression and neural-network models (best RMSE 0.30 lb) to select motor settings
 - Optimized settings predicted within 0.05 lb of every 1-8 lb target weight

AR4 6-Axis Robotic Arm - Systems Design: Designed and built a 6-axis arm on a Teensy 4.1 with a custom motor-control cabinet and PLC-triggered motion sequences

CERTIFICATIONS

FANUC Academy HandlingTool Operations & Programming (May '26) | Beckhoff TwinCAT 3 Getting Started (May '24) | FANUC CERT Instructor Certification, FANUC America (in progress)

EDUCATION

B.S. Mechatronics Engineering; A.S. Pre-Engineering (May '23)

Utah Tech University

Saint George, UT